

DATA VORTEX 2026

ROUND 1 - PHASE 2

LOGIC EXPLANATION

Analytical reasoning, validation, and assumptions

Prepared from the project SQLite database using read-only SELECT analysis.

EASY - E2 - Most Engaged Posts

Official question: Find the top 10 posts based on total engagement. Total engagement is defined as likes + shares + comments. Ignore posts where likes are missing.

Analytical objective

Identify the ten complete-engagement posts with the highest combined interactions.

Data and filtering

The posts table supplies the post ID, platform, and interaction columns. The official instruction requires missing likes to be ignored, so WHERE likes IS NOT NULL is applied before ranking.

Calculation and ordering

Total engagement is calculated row by row as likes + shares + comments. Ordering descending returns the largest totals; post_id is a deterministic tie-breaker.

Validation and edge cases

The query returned exactly 10 rows, all with non-null likes. No joins are used, so there is no risk of duplicate multiplication.

Executed result: **10 row(s)** returned.

MEDIUM - M1 - Which Locations Generate the Most Engagement?

Official question: Using both datasets, calculate the total engagement generated by users from each location. Return the location, number of posts, and total engagement. Rank locations from highest to lowest engagement.

Analytical objective

Measure the total complete engagement associated with creators at each location.

Tables and join

users provides location and posts provides engagement. The one-to-many join uses `posts.user_id = users.user_id`, the database foreign-key relationship.

Filtering and aggregation

Posts with missing likes are excluded because a complete likes + shares + comments total cannot be calculated. GROUP BY location then produces the requested post count and total engagement.

Validation and edge cases

Grouped post counts reconcile to the number of posts with non-null likes, and grouped engagement totals reconcile exactly to a direct database-wide total. Every retained post has one matching user.

Executed result: **33 row(s)** returned.

HARD - H2 - Rank Users Within Their Location

Official question: For every location, rank users based on their total engagement. Return only the top 3 users from each location.

Analytical objective

Rank creators within their own locations, then retain the leading three positions per location.

Transformation

The first CTE creates one row per user after joining users to posts and aggregating complete engagement. This prevents a creator with many posts from appearing multiple times in the ranking stage.

Window ranking

DENSE_RANK partitions by location and orders total engagement descending. The final filter retains engagement_rank <= 3, preserving any genuine ties at a top-three rank.

Validation and edge cases

All output ranks are between 1 and 3. The returned row count reconciles to the sum of the first three available ranking positions across locations; the read-only analysis avoids source-database mutation.

Executed result: **99 row(s)** returned.